XRPL Validator — Daily Engineering Report · 2026-11-12

Period: 2026-11-12 00:00 to 24:00 US Eastern (2026-11-12 05:00 to 2026-11-13 05:00 UTC); ledgers #108,275,272 to #108,296,676. Every count from the ledger and the validator's logs stops at #108,296,676, closed 04:59:51 UTC. Generated 2026-11-13 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 100.000% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,405 of the 21,405 ledgers the network closed (100.000%); every validation matched the network's validated ledger.
- **Native engine:** 0 engine receipts.
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 8 slow-close periods; the longest, 14:42 UTC–14:42 UTC, closed ledgers up to 20 s apart.
- **Independence program:** 15 of 30 qualifying days (snapshot at 05:20:42 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,405
Validations issued	21,405
Coverage (validations / ledgers closed)	100.000%
Validations matching the network's validated ledger hash	21,405 of 21,405
Unsigned during planned restarts	0
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%
Restarts	0
Basis: our own ledger hash	21,405
External agreement, data.xrpl.org (UTC day 2026-11-12)	1.00000 (0 missed of 20,609)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,275,272–#108,296,676), by step:

Step	Ledgers with a receipt
Step 1 · native engine	0
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters, snapshot at 05:20:42 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	172,196 ledgers, 35,418,979 transactions	172,194 fully matched; results 2, bytes 64, keys 1/12
2 · transaction records	175,966 ledgers, 36,232,654 records	36,232,394 exact; differ 256, tree 3
3 · transaction order	176,009 ledgers (mode: on)	176,004 exact; order 5, results 1, bytes 0, refused 5
4 · ledger data from peers	176,015 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	175,955 ledgers judged	175,922 agreed; differ 0, short of quorum 33
5 · our own ledger hash	176,013 ledgers	175,957 own; differ 3, partial 53
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

No receipts this day.

Incidents

No incidents.

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **15 of 30** (snapshot at 05:20:42 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-12 — tracked #108,270,742–#108,292,245	Value
Validations	21,504 (21,504 own, 0 borrowed)
Coverage of the tracked ledgers	100.00%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,405 (#108,275,272–#108,296,676)
Mean close interval	4.04 s
Longest close interval	20 s (before #108,284,575)
Closes at least 15 s apart	8
Empty ledgers	0

- Slow closes 05:34:02 UTC–05:34:20 UTC: ledgers #108,275,792, 1 close at least 15 s apart, up to 18 s.
- Slow closes 14:42:30 UTC–14:42:50 UTC: ledgers #108,283,966, 1 close at least 15 s apart, up to 20 s.
- Slow closes 14:48:21 UTC–14:48:40 UTC: ledgers #108,284,045, 1 close at least 15 s apart, up to 19 s.
- Slow closes 15:01:22 UTC–15:01:40 UTC: ledgers #108,284,232, 1 close at least 15 s apart, up to 18 s.
- Slow closes 15:25:40 UTC–15:26:00 UTC: ledgers #108,284,575, 1 close at least 15 s apart, up to 20 s.
- Slow closes 02:56:02 UTC–02:56:20 UTC: ledgers #108,294,870, 1 close at least 15 s apart, up to 18 s.
- Slow closes 04:15:22 UTC–04:15:40 UTC: ledgers #108,296,021, 1 close at least 15 s apart, up to 18 s.
- Slow closes 04:55:51 UTC–04:56:10 UTC: ledgers #108,296,622, 1 close at least 15 s apart, up to 19 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

None.

Corrections to earlier reports

None.